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Introduction to Blockchain Technology

Lesson 6 – Bitcoin Mining, Proof of Work, and Incentives



Bitcoin Mining, Proof of Work, and Incentives



- 1. Mining and Block Creation**
- 2. Bitcoin Block and Block Header**
- 3. Difficulty Target**
- 4. Proof of Work (Bitcoin Consensus)**
- 5. Miner Rewards and Incentives**
- 6. Security and Limitations**
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1. Mining and Block Creation

A Bitcoin block contains:

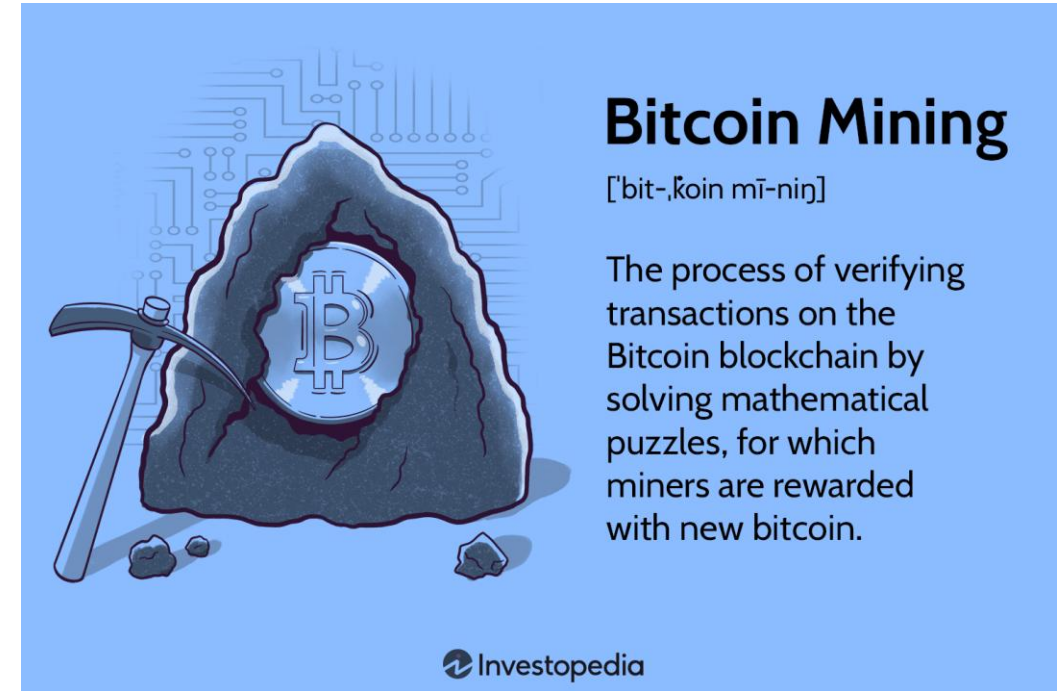
- ❖ Block header (metadata)
- ❖ Block body: List of transactions

Mining repeatedly hashes the **block header** to find a valid block hash.

1. Mining and Block Creation

Mining is the process that:

- ❖ Groups transactions into blocks
- ❖ Determines the order of transactions
- ❖ Adds new blocks to the blockchain
- ❖ Secures the network against attacks



Mining allows a decentralized network to agree on one shared history.

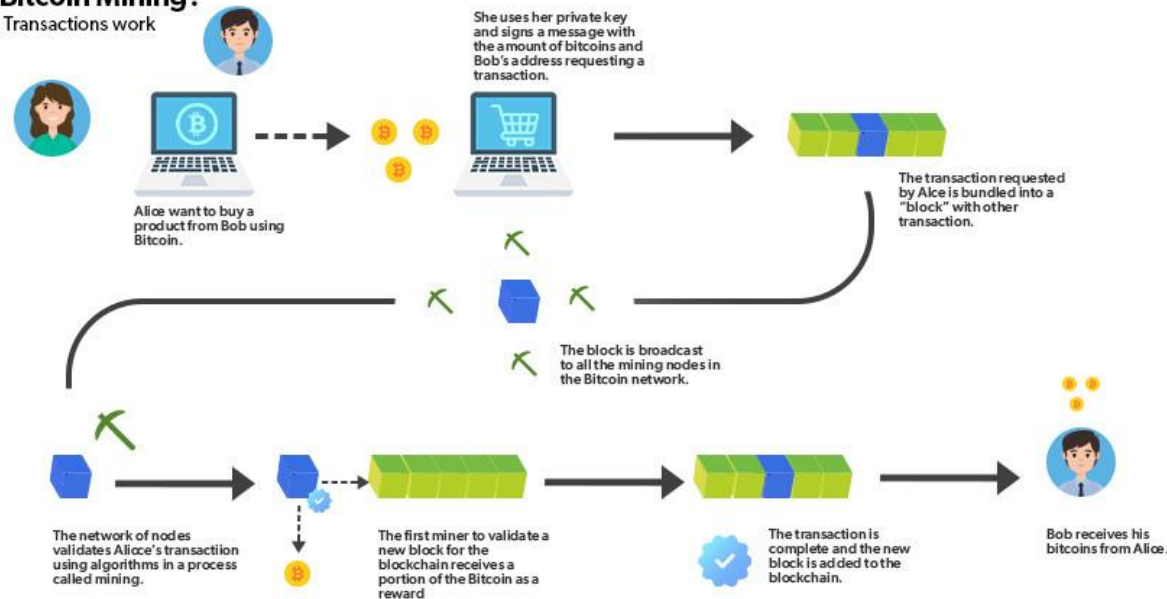
1. Mining and Block Creation

Mining introduces:

- ❖ Computational cost
- ❖ Economic incentive
- ❖ Difficulty in rewriting history

What is Bitcoin Mining?

How Bitcoin Transactions work



Bitcoin uses **computational cost** instead of **central permission** to control block creation.

2. Bitcoin Block and Block Header

Bitcoin Block



Bitcoin Block 931,723

Mined on January 10, 2026 09:43:42 • [All Blocks](#)

AntPool

Coinbase Message •

HMined by AntPool G +*DO'Lz>mmD9+4N+C*abS1 yb3?M+?WJ/1_rOEc3G+ I'A r

A total of 3,993.23 BTC (\$361,158,912) were sent in the block with the average transaction being 1.1528 BTC (\$104,262). AntPool earned a total reward of 3.13 BTC \$283,085. The reward consisted of a base reward of 3.13 BTC \$283,085 with an additional 0.0256 BTC (\$2,315.33) reward paid as fees of the 3,464 transactions which were included in the block.

Details

Hash	00000-03b4a	Depth	1
Capacity	152.32%	Size	1,597,170
Distance	40m 51s	Version	0x20010000
BTC	3,993.2319	Merkle Root	5f-e1
Value	\$361,158,912	Difficulty	146,472,570,619,930.78
Value Today	\$361,204,595	Nonce	1,779,521,350
Average Value	1.1527805657 BTC	Bits	386,001,906
Median Value	0.00428597 BTC	Weight	3,994,077 WU
Input Value	3,993.26 BTC	Minted	3.13 BTC
Output Value	3,996.38 BTC	Reward	3.15056598 BTC
Transactions	3,464	Mined on	Jan 10, 2026, 9:43:42 PM
Witness Tx's	3,212	Height	931,723
Inputs	7,862	Confirmations	1
Outputs	9,069	Fee Range	0-100 sat/vByte
Fees	0.02556598 BTC	Average Fee	0.00000738
Fees Kb	0.0000160 BTC	Median Fee	0.00000288
Fees kWU	0.0000064 BTC	Miner	AntPool

Transactions

↕	Last	First	↗ Value	↘ Value
↗ Fee	↘ Fee			
TX	0 ID: fc9f-5659	From Block Reward	3.15056598 BTC • \$284,945	Fee 0 Sats • \$0.00
	1/10/2026, 21:43:42	To 7 Outputs		
TX	1 ID: e6d0-40c5	From bc1q-xx9w	0.15945336 BTC • \$14,421.40	Fee 36.8K Sats • \$33.24
	1/10/2026, 21:33:54	To 3 Outputs		
TX	2 ID: b5fe-d144	From 38Ek-f4dr	0.19824000 BTC • \$17,929.37	Fee 176.0K Sats • \$159.18
	1/10/2026, 21:40:39	To 44 Outputs		
TX	3 ID: 8f2b-5dca	From 38n3-exa9	0.01477300 BTC • \$1,336.11	Fee 9.9K Sats • \$8.95
	1/10/2026, 21:28:05	To 1AR9-ynHM		
TX	4 ID: 73ef-d0de	From bc1q-cyu2	0.00995930 BTC • \$900.75	Fee 9.9K Sats • \$8.93
	1/10/2026, 21:28:56	To 2 Outputs		
TX	5 ID: 20a6-d46c	From bc1q-vzej	0.00802642 BTC • \$725.93	Fee 11.5K Sats • \$10.42
	1/10/2026, 21:33:14	To 2 Outputs		
TX	6 ID: 6516-523e	From 6 Inputs	0.19532400 BTC • \$17,665.64	Fee 29.7K Sats • \$26.89
	1/10/2026, 21:43:42	To 5 Outputs		
TX	7 ID: fc35-6746	From bc1q-30yn	0.01411387 BTC • \$1,276.50	Fee 7.0K Sats • \$6.38
	1/10/2026, 21:34:24	To 2 Outputs		
TX	8 ID: dc7d-1438	From bc1q-hchp	0.01069460 BTC • \$967.25	Fee 7.0K Sats • \$6.38
	1/10/2026, 21:43:37	To 2 Outputs		

2. Bitcoin Block and Block Header

Block Header contains:

- ❖ **Version:** A number indicating the set of validation rules to follow
- ❖ **Previous block hash:** Hash of the previous block's header
- ❖ **Merkle root:** A single hash summarizing all transactions in the block
- ❖ **Timestamp:** Approximate creation time of the block
- ❖ **Difficulty target:** A value that the block hash must be smaller than
- ❖ **Nonce:** A number controlled by the miner
- ❖ Other consensus-related fields

Only the block header is repeatedly hashed during mining.

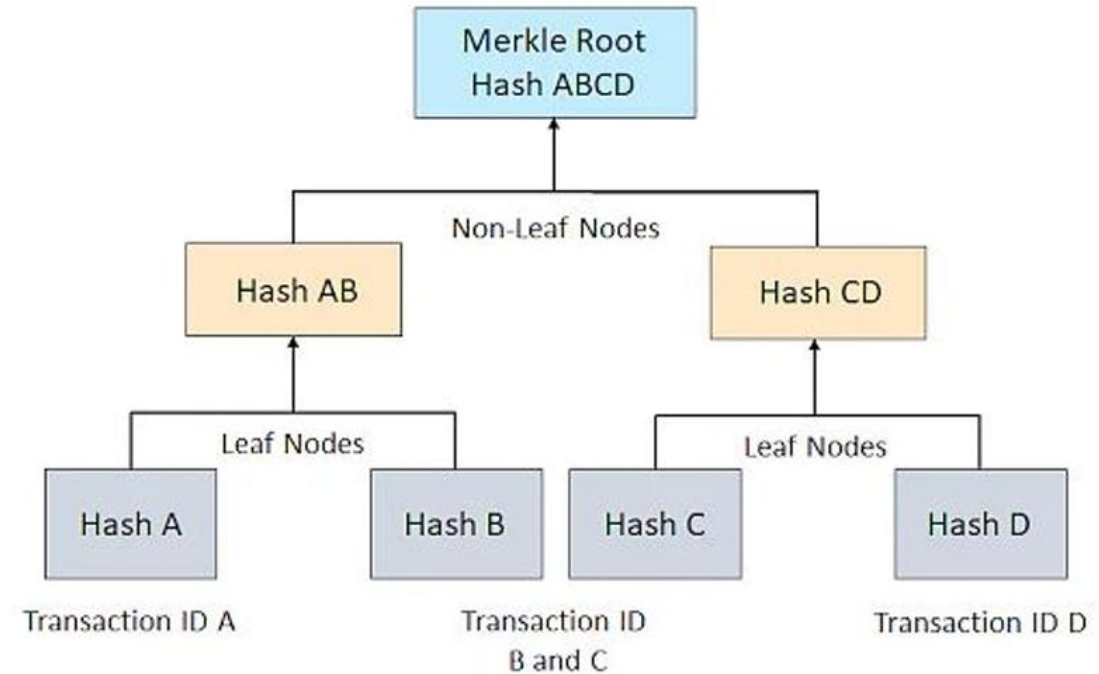
2. Bitcoin Block and Block Header

Merkle Root:

- ❖ A single hash representing all transactions.
- ❖ Any change in a transaction changes the Merkle Root, which changes the block hash.

Steps:

1. Take each pair of Transaction IDs from the block, and hash through SHA256 twice. them together
2. Keep doing this for each pair Transaction IDs, until you end up with a new list of hashes.
3. If you have an odd number of transactions, hash the remaining transaction with itself.
4. Repeat steps 1-2 for every new list of hashes you create until you finally end up with one hash



2. Bitcoin Block and Block Header

Nonce

- ❖ 32-bit integer that is included in the block
- ❖ The nonce is hashed together with the block header fields and to create the hash for the entire block
- ❖ Controlled by the miner
- ❖ Allows miners to change the block hash without changing transactions
- ❖ Changing the nonce changes the hash result.

Q: Will changing the nonce affect transactions?

Answer: No, Because nonce is part of the block header, not the transaction list.

3. Difficulty Target

What is Difficulty Target?

- ❖ Define how hard mining is
- ❖ Regulate how long it takes for miners to add new blocks of transactions to the blockchain.
- ❖ It is a threshold value for the block hash
- ❖ A block is valid if ***Block Hash < Difficulty Target***
 - Lower target → harder mining
 - Higher target → easier mining
- ❖ The difficulty target is not chosen by miners
- ❖ It is enforced by all full nodes

Q: Who determines the difficulty target in Bitcoin?

Answer: Every full node independently calculates it using protocol rules.

3. Difficulty Target

The Mining Loop: Searching for the Threshold

- ❖ Miners take the Block Header.

- ❖ Goal: Find a Nonce such that:

$$SHA-256(SHA-256(Block\ Header + Nonce)) < Difficulty\ Target$$

- ❖ **Brute Force:** Because hashing is unpredictable, there is no way to calculate the nonce directly. Miners must try trillions of random numbers until they find one that works

- ❖ The Merkle Root and Previous Hash are **Fixed**. Only the Nonce and Timestamp are **changed** to find a new hash

3. Difficulty Target

Difficulty Adjustment:

- ❖ Bitcoin adjusts difficulty every 2016 blocks.
 - It ensures that blocks of transactions are added to the blockchain at regular intervals, even as more miners join the network.
 - If the difficulty remained the same, it would take less time between adding new blocks to the blockchain as new miners join the network. Expected Block time $\approx$ 10 minutes

Q: What is the design purpose of Bitcoin when adjusting difficulty target?

3. Difficulty Target

Example: A miner finds a block with the following data:

Difficulty Target: **00000000FFFF...**

Calculated Block Hash: **00000001AAAA...**

Tasks:

- Is this block valid? Explain your answer?
- If the miner wants to fix it, which specific fields in the header can they change?

4. Proof of Work (Bitcoin Consensus)

Mostly fixed during mining

- ❖ Previous block hash
- ❖ Merkle root
- ❖ Difficulty target
- ❖ Version

Adjustable

- ❖ Nonce
- ❖ Timestamp (only changes occasionally during mining process)

Only adjustable fields allow miners to search for a valid hash.

4. Proof of Work (Bitcoin Consensus)

Mining Process: Only repeated hashing

- ❖ Miner builds a candidate block
- ❖ Sets nonce to an initial value
- ❖ Hashes the block header
- ❖ Checks if hash < difficulty target

- ❖ If not:
 - Change nonce
 - Try again

4. Proof of Work (Bitcoin Consensus)

Example: Say that Nam sends An 10 BTC

- ❖ It gets added to a block, along with other transactions, which is hashed by the BTC network into *ABC...XYZ*
- ❖ The crypto network says: find the Nonce that when appended to the block, produces a hash with less than *EFG...KLM*.
- ❖ Miner hashes block header many times. Each hash is unpredictable. Success occurs when $\text{hash} < \text{difficulty target}$. Once it is found, the block is validated and gets added to the network
- ❖ The block cannot be validated until the nonce can be found.
- ❖ Once the correct hash is found, the transaction and the hash are permanently stored in the blockchain, and if anyone tries to change the information in the block, the hashes will mismatch.

4. Proof of Work (Bitcoin Consensus)

Why Proof of Work Is Secure

- ❖ Finding a valid block is hard
- ❖ Verifying a block is easy
- ❖ Attacking the network requires massive computation

⇒ Attacks are possible but economically expensive.

Q: Why is Proof of Work hard to perform but easy to verify?

5. Miner Rewards and Incentives

Coinbase transaction

- ❖ A special transaction created by the miner
- ❖ The first transaction in every block
- ❖ Has:
 - No inputs
 - Only outputs

The miner gets paid for:

- ❖ Block reward (defined by Bitcoin network)
- ❖ All transaction fees in the block
- ❖ Transaction fees do NOT create UTXOs directly
- ❖ Fees are collected here

5. Miner Rewards and Incentives

Coinbase transaction: Creating New Supply

- ❖ Unlike normal transactions (which spend UTXOs), the Coinbase transaction has no inputs.
 - It is the only way the system makes new money out of nothing
- ❖ This transaction pays the miner two things: the New Coin Reward plus all the transaction fees paid by the people in that block
- ❖ **Waiting Rule:** The miner cannot spend these new coins immediately. They must wait for 100 more blocks to be added to the chain first. This rule protects the system if there is a fork

5. Miner Rewards and Incentives

Block Subsidy (Block Reward)

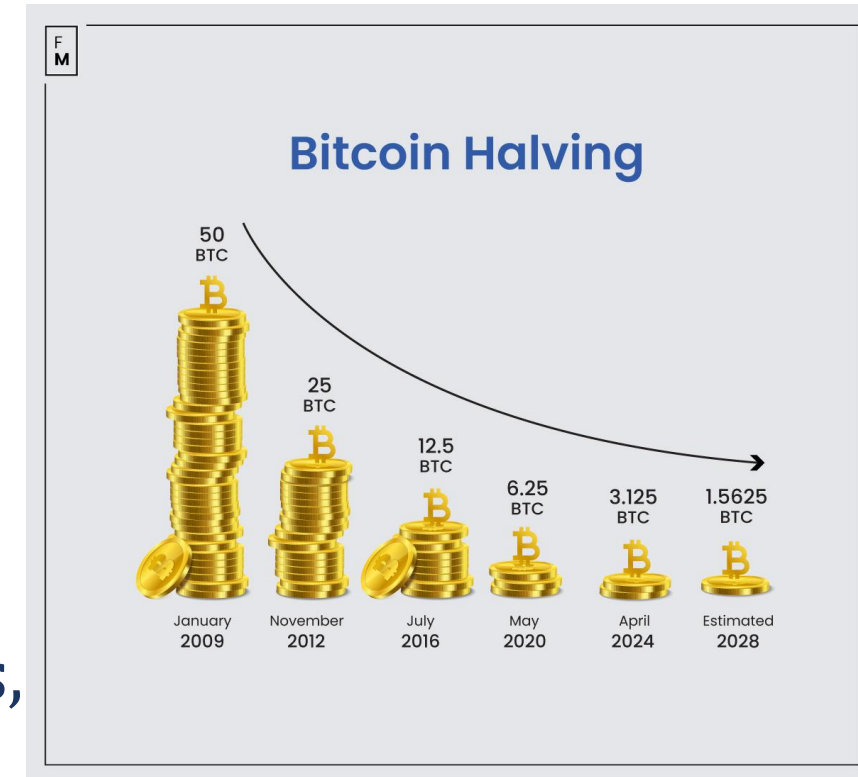
- ❖ Block subsidy is newly created bitcoin
- ❖ It is defined by the Bitcoin protocol
- ❖ Paid through the coinbase transaction
- ❖ Halves every 210,000 blocks (~4 years)
- ❖ Controls Bitcoin's total supply (21 million BTC)

Block subsidy is how Bitcoin distributes new coins, and **halving** is how Bitcoin controls inflation.

Over time, fees become more important than subsidy

Bitcoin Subsidy as of November 2025: 3.125 BTC per block

Q: What happens when block subsidy becomes very small?



Subsidy approaches zero, and never increases.

5. Miner Rewards and Incentives

How Transaction Fees Reach the Miner

- ❖ Miner calculates sum of all transaction fees in the block
- ❖ Miner adds them to the coinbase transaction output

Example

Block reward: 3.125 BTC

Total fees in block: 0.5 BTC

Coinbase output = 3.625 BTC → miner

Fees become spendable **only after block confirmation.**

6. Security and Limitations

Security Provided by Proof of Work

- ❖ Altering a block requires redoing Proof of Work
- ❖ Rewriting history requires controlling most hash power
- ❖ Verification is fast and independent for each node
- ❖ Economic cost discourages attacks

=> Security comes from **computation + incentives**, not trust

6. Security and Limitations

Limitations of Proof of Work

- ❖ High energy consumption
- ❖ Low transaction throughput
- ❖ Mining centralization risk
- ❖ 51% attacks are theoretically possible

Attacks are possible, but **economically expensive**

Q: Why does Bitcoin accept high energy cost?

7. Summary

- ❖ Mining groups transactions into blocks
- ❖ Only the block header is hashed during mining
- ❖ Proof of Work secures consensus through computation
- ❖ Difficulty target keeps block time stable
- ❖ Miners are rewarded via subsidy and transaction fees
- ❖ Bitcoin stays secure because miners earn incentives by following the network rules

8. Discussion

1. If two miners find valid blocks at the same time, is the system broken?
2. Why are transaction fees important for Bitcoin's future security?
3. Would lowering difficulty permanently make Bitcoin more efficient?
4. As of January 2026, a miner creating a block contains the following 3 transactions, aside a coinbase transaction:

Tx	Total Inputs (BTC)	Total Outputs (BTC)
Tx1	5.0	4.8
Tx2	2.5	2.4
Tx3	1.2	1.0

The coinbase transaction has one output of 3.9 BTC.

- a. Calculate the total transaction fees in this block.
- b. Is this block valid? Why or why not?